# A CANONICAL MAP FROM THE CHOW GROUP OF K-CYCLES TO K_0 OF THE QUOTIENT OF COHERENT SHEAVES IN ADJACENT DIMENSIONS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

*Allowed standard prerequisites (editorial).*

- Isomorphism between k-cycles and K_0 of the truncated coherent category. Same chapter chow.tex, label lemma-cycles-k-group, invoked at proof line 3944. Provides the map from Z_k(X) to K_0(Coh_{<=k}(X)/Coh_{<=k-1}(X)) whose composite with the quotient map is the asserted canonical map; the descent through rational equivalence is what the counted proof supplies.
- Pushforward on K_0 for a finite morphism of schemes. Same chapter chow.tex, label lemma-finite-cycles-k-group, invoked at proof line 3959. Lets the disjoint union W of the integral subschemes W_j be used in place of X, so that one may assume X integral of dimension k + 1.
- The ideal of denominators of a rational function. Divisors, labels divisors-definition-regular-meromorphic-ideal-denominators (proof line 3971) and divisors-lemma-regular-meromorphic-ideal-denominators (proof line 3982). Produces the ideal I with the two short exact sequences whose difference of classes represents div(f), which is the construction at the heart of the proof.
- The distance function and the order of vanishing of a determinant. Algebra, labels algebra-definition-distance (proof line 4008), algebra-lemma-properties-distance-function and algebra-lemma-order-vanishing-determinant (proof lines 4014-4015); same chapter chow.tex, label definition-rational-equivalence (proof line 3952). Converts length_A(A/I) - length_A(A/fI) into ord_A(f) so that the claimed cycle identity matches the coefficient of div(f).

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*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 02MG: sources/stacks-chow/tags.txt, source lines 3443–3443. Navigation pointer for \ref{algebra-definition-distance} used inside the counted span (line 4008); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02MI: sources/stacks-chow/tags.txt, source lines 3445–3445. Navigation pointer for \ref{algebra-lemma-order-vanishing-determinant} used inside the counted span (line 4015); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02MH: sources/stacks-chow/tags.txt, source lines 3444–3444. Navigation pointer for \ref{algebra-lemma-properties-distance-function} used inside the counted span (line 4014); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02RW: sources/stacks-chow/pinned-original-source.tex, source lines 2882–2882. Navigation pointer for \ref{definition-rational-equivalence} used inside the counted span (line 3952); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02P1: sources/stacks-chow/tags.txt, source lines 3536–3536. Navigation pointer for \ref{divisors-definition-regular-meromorphic-ideal-denominators} used inside the counted span (line 3971); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02P0: sources/stacks-chow/tags.txt, source lines 3535–3535. Navigation pointer for \ref{divisors-lemma-regular-meromorphic-ideal-denominators} used inside the counted span (line 3982); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02S9: sources/stacks-chow/pinned-original-source.tex, source lines 3757–3757. Navigation pointer for \ref{lemma-cycles-k-group} used inside the counted span (line 3944); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 0FDR: sources/stacks-chow/pinned-original-source.tex, source lines 3898–3898. Navigation pointer for \ref{lemma-finite-cycles-k-group} used inside the counted span (line 3959); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3936); the argument that consumes it is retained in the span.

**Lemma 0.1.** *Let $X$ be a scheme locally of finite type over $(S, \delta)$ as in Situation 02QL. There is a canonical map*

$$\mathrm{CH}_k(X) \longrightarrow K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$$

*induced by the map $Z_k(X) \to K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X))$ from Lemma 02S9.*

*Proof.* We have to show that an element $\alpha$ of $Z_k(X)$ which is rationally equivalent to zero, is mapped to zero in $K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$. Write $\alpha = \sum (i_j)_* \mathrm{div}(f_j)$ as in Definition 02RW. Observe that

$$\pi = \coprod i_j : W = \coprod W_j \longrightarrow X$$

is a finite morphism as each $i_j : W_j \to X$ is a closed immersion and the family of $W_j$ is locally finite in $X$. Hence we may use Lemma 0FDR to reduce to the case of $W$. Since $W$ is a disjoint union of integral scheme, we reduce to the case discussed in the next paragraph.

Assume $X$ is integral of $\delta$-dimension $k+1$. Let $f$ be a nonzero rational function on $X$. Let $\alpha = \mathrm{div}(f)$. We have to show that $\alpha$ is mapped to zero in $K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$. Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal of denominators of $f$, see Divisors, Definition 02P1. Then we have short exact sequences

$$0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I} \to 0$$

and

$$0 \to \mathcal{I} \xrightarrow{f} \mathcal{O}_X \to \mathcal{O}_X/f\mathcal{I} \to 0$$

See Divisors, Lemma 02P0. We claim that

$$[\mathcal{O}_X/\mathcal{I}]_k - [\mathcal{O}_X/f\mathcal{I}]_k = \mathrm{div}(f)$$

The claim implies the element $\alpha = \mathrm{div}(f)$ is represented by $[\mathcal{O}_X/\mathcal{I}] - [\mathcal{O}_X/f\mathcal{I}]$ in $K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X))$. Then the short exact sequences show that this element maps to zero in $K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$.

To prove the claim, let $Z \subset X$ be an integral closed subscheme of $\delta$-dimension $k$ and let $\xi \in Z$ be its generic point. Then $I = \mathcal{I}_\xi \subset A = \mathcal{O}_{X,\xi}$ is an ideal such that $fI \subset A$. Now the coefficient of $[Z]$ in $\mathrm{div}(f)$ is $\mathrm{ord}_A(f)$. (Of course as usual we identify the function field of $X$ with the fraction field of $A$.) On the other hand, the coefficient of $[Z]$ in $[\mathcal{O}_X/\mathcal{I}] - [\mathcal{O}_X/f\mathcal{I}]$ is

$$\mathrm{length}_A(A/I) - \mathrm{length}_A(A/fI)$$

Using the distance function of Algebra, Definition 02MG we can rewrite this as

$$d(A, I) - d(A, fI) = d(I, fI) = \mathrm{ord}_A(f)$$

The equalities hold by Algebra, Lemmas 02MH and 02MI. (Using these lemmas isn't necessary, but convenient.) $\square$